

Influence Functions

Empirical Distribution Function

Definition 1.1. Suppose $(X_i) \stackrel{\text{IID}}{\sim} F_X$. The **empirical (cumulative) distribution function** $\mathbb{F}_n$ is defined as:

$$\mathbb{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x).$$

■

Definition 1.2. The **Kolmogorov-Smirnov** distance between two distributions functions F and G , mapping $\mathbb{R} \rightarrow \mathbb{R}$, is defined as:

$$d(F, G) = \sup_{x \in \mathbb{R}} |F(x) - G(x)|. \quad (1.1)$$

■

Theorem 1.1 (Glivenko-Cantelli). Suppose $(X_i) \stackrel{\text{IID}}{\sim} F_X$. The empirical distribution function $\mathbb{F}_n$ converges *uniformly almost surely* to F_X :

$$\sup_{x \in \mathbb{R}} |\mathbb{F}_n(x) - F_X(x)| \xrightarrow{as} 0.$$

□

Remark 1.1. See Serfling (1980), theorem 2.1.4.

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Statistical Functionals

2.1 Definition

Definition 2.1. A **functional** T is a mapping from a function space $\mathcal{F}$ into $\mathbb{R}$. Typically, $\mathcal{F}$ is a linear function (vector) space.

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Example 2.1. Let $\{F, F_0\}$ denote distribution functions. Examples of *statistical functionals* include:

- The expectation of a specified function $\psi : \mathbb{R} \rightarrow \mathbb{R}$:

$$T(F) = \int \psi(x) dF(x). \quad (2.1)$$

Functionals of the form (2.1) are **linear**.

- The k th central moment:

$$T(F) = \int \left\{ x - \int \xi dF(\xi) \right\}^k dF(x).$$

- The *Cramer von Mises* distance:

$$T(F) = \int \{F(x) - F_0(x)\}^2 dF_0(x).$$



Example 2.2. Let H_x denote the Heaviside function:

$$H_x(t) = \int_{-\infty}^t \delta_x(u) du = \mathbb{I}(t \geq x),$$

where $\delta_x(u)$ is a Dirac spike localized to x . The empirical distribution function is expressible as:

$$\mathbb{F}_n(t) = \frac{1}{n} \sum_{i=1}^n H_{X_i}(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq t).$$

The evaluation of a linear functional at $\mathbb{F}_n$ is:

$$T(\mathbb{F}_n) = \int \psi(x) d \left\{ \frac{1}{n} \sum_{i=1}^n H_{X_i}(x) \right\} = \frac{1}{n} \sum_{i=1}^n \int \psi(x) \delta_{X_i}(x) dx = \frac{1}{n} \sum_{i=1}^n \psi(X_i).$$



2.2 Continuity

Definition 2.2 (Lehmann 1999, definition 6.2.1). Let d denote the KS metric (1.1). A statistical functional T is **continuous** if for any sequence $G_n \rightarrow F$,

$$\lim_{n \rightarrow \infty} d(G_n, F) = 0 \implies \lim_{n \rightarrow \infty} T(G_n) = T(F). \quad (2.2)$$



Proposition 2.1 (Functional Consistency). Suppose $(X_i) \stackrel{\text{IID}}{\sim} F_X$, and T is a continuous functional (2.2), then $T(\mathbb{F}_n)$ is consistent for $T(F)$, i.e. $T(\mathbb{F}_n) \xrightarrow{p} T(F)$. $\blacklozenge$

Proposition 2.2 (Huber 2009, lemma 2.1). A linear functional L defined on the space $\mathcal{F}$ of probability measures on $(\mathbb{R}, \mathcal{B})$ is **continuous** if and only if it is expressible as:

$$L(F) = \int \psi(x) dF(x),$$

for some *bounded, continuous* function $\psi : \mathbb{R} \rightarrow \mathbb{R}$. $\blacklozenge$

2.3 Differentiability

Definition 2.3. Let F and G denote distribution functions. Define the **ϵ -contaminated** distribution:

$$F_{G,\epsilon} = (1 - \epsilon)F + \epsilon G. \quad (2.3)$$

The **Gateaux differential** $\partial T(F; G - F)$ of the functional T at F in the direction of $G - F$ is defined by:

$$\partial T(F; G - F) = \lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} \{T(F_{G,\epsilon}) - T(F)\}. \quad (2.4)$$

In general, the k th Gateaux differential is obtained via:

$$\partial^k T(F; G - F) = \lim_{\epsilon \downarrow 0} \left\{ \frac{\partial^k T(F_{G,\epsilon})}{\partial \epsilon^k} \right\}.$$

■

Example 2.3. Consider the Cramer von Mises functional:

$$T(F) = \int (F - F_0)^2 dF_0.$$

The evaluation at the ϵ -contaminated distribution is:

$$T(F_{G,\epsilon}) = \int (F + \epsilon(G - F) - F_0)^2 dF_0.$$

The derivative with respect to ϵ is:

$$\frac{\partial}{\partial \epsilon} T(F_{G,\epsilon}) = 2 \int (F + \epsilon(G - F) - F_0)(G - F) dF_0$$

Taking the limit as $\epsilon \downarrow 0$ gives:

$$\partial T(F; G - F) = \lim_{\epsilon \downarrow 0} \frac{\partial}{\partial \epsilon} T(F_{G,\epsilon}) = 2 \int (F - F_0)(G - F) dF_0.$$

♠

Discussion 2.1. A functional T is **Frechet differentiable** if there exists a functional $\partial T(F; G - F)$, *linear in $G - F$* , such that:

$$|T(G_n) - T(F) - \partial T(F; G_n - F)| = o\{d(F, G_n)\}$$

for all sequences (G_n) such that $d(G_n, F) \rightarrow 0$ as $n \rightarrow \infty$.

If T is continuous in a neighborhood of F and *Frechet differentiable* at F , then the Frechet derivative $\partial T(F; G - F)$ is *continuous* at F (Huber 2009, proposition 2.19). By proposition (2.2), there exists a bounded, continuous function ψ_F such that:

$$\partial T(F; G - F) = \int \psi_F(x) d\{G(x) - F(x)\} = \int \left\{ \psi_F(x) - \int \psi_F(\xi) dF(\xi) \right\} dG(x).$$

Moreover, if the Frechet derivative exists, then the Gateaux derivative exists, and the two are equal (Serfling 1980, section 6.2). Thus, continuity and Frechet differentiability at F suffice to represent the Gateaux derivative $\partial T(F; G - F)$ as:

$$\partial T(F; G - F) = \int \varphi_F(x) dG(x),$$

where φ_F is bounded, continuous, and mean-zero. When Frechet differentiability cannot be established, it is often *assumed* that the Gateaux derivative is expressible as:

$$\partial h(F; G - F) = \int \psi_F(x) d\{G(x) - F(x)\} = \int \varphi_F(x) dG(x).$$

for some measurable function ψ_F . Bounding and continuity of $\varphi_F(x)$ are not assumed. ♠

Influence Functions

3.1 Definition

Example 3.1. Suppose T is a statistical functional whose Gateaux derivative admits the representation:

$$\partial T(F; G - F) = \int \varphi_F(x) dG(x).$$

To isolate the function $\varphi_F(x)$, set $G(t) = H_x(t)$, then:

$$\partial T(F; H_x - F) = \int \varphi_F(t) dH_x(t) = \int \varphi_F(t) \delta_x(t) dt = \varphi_F(x).$$

Therefore, for any other G , the Gateaux derivative is expressible as:

$$\partial T(F; G - F) = \int \varphi_F(x) dG(x) = \int \{\partial T(F; H_x - F)\} dG(x).$$

$\varphi_F = \partial T(F; G - F)$ is described as the *influence function*. ♠

Definition 3.1. Suppose $(X_i) \stackrel{\text{iid}}{\sim} F$, and T is a statistical functional. Recall that $H_x(t) = \mathbb{I}(x \leq t)$ denotes the Heaviside function. The **influence function** of T is the Gateaux derivative of T at F in the direction of $H_x - F$:

$$\partial T(F; H_x - F) = \lim_{\epsilon \downarrow 0} \left[\frac{\partial}{\partial \epsilon} T\{F + \epsilon(H_x - F)\} \right]. \quad (3.1)$$

■

3.2 Properties

Proposition 3.1. Influence functions inherit the chain, product, and quotient rules from differentiation.

i. **(Linearity)** If $T = \sum_{j=1}^m \alpha_j T_j$ is a linear combination of functionals, then:

$$\partial T(F; H_x - F) = \sum_{j=1}^m \alpha_j \partial T_j(F; H_x - F).$$

ii. **(Product Rule)** If $T = T_1 \cdot T_2$ is a product of functionals, then:

$$\partial T(F; H_x - F) = \partial T_1(F; H_x - F) \cdot T_2(F) + T_1(F) \cdot \partial T_2(F; H_x - F).$$

iii. **(Quotient Rule)** If $T = T_1/T_2$ is a ratio of functionals, then:

$$\partial T(F; H_x - F) = \frac{\partial T_1(F; H_x - F) \cdot T_2(F) - T_1(F) \cdot \partial T_2(F; H_x - F)}{T_2^2(F)}.$$

iv. **(Chain Rule)** If $T = U \circ T_1$ is a differentiable function of a functional, then:

$$\partial T(F; H_x - F) = \dot{U}\{T_1(F)\} \partial T_1(F; H_x - F).$$

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Example 3.2 (Linear Functional). Consider the linear functional:

$$T(F) = \int \psi(x) dF(x).$$

Evaluating at $F_{x,\epsilon}$:

$$T(F_{x,\epsilon}) = \int \psi(u) d\{(1 - \epsilon)F(u) + \epsilon H_x(u)\} = (1 - \epsilon) \int \psi(u) dF(u) + \epsilon \psi(x)$$

Taking the derivative with respect to ϵ :

$$\frac{\partial T(F_{x,\epsilon})}{\partial \epsilon} = - \int \psi(u) dF(u) + \psi(x).$$

Thus, the influence function is:

$$\varphi_F(x) = \psi(x) - \mathbb{E}\{\psi(x)\}.$$



Example 3.3 (Sample Variance). The variance functional is expressible as:

$$T(F) = \int \{x - T_1(F)\}^2 dF(x),$$

where $T_1(F)$ is the mean functional. Expanding the quadratic:

$$T(F) = \int \{x^2 + T_1^2(F) - 2xT_1(F)\} dF(x) = T_2(F) - T_1^2(F).$$

where T_2 is the second moment functional. Since the k th moment is a linear functional, the influence functions for T_1 and T_2 are:

$$\varphi_1(x) = x - \mu_1, \quad \varphi_2(x) = x^2 - m_2.$$

By linearity and the chain rule, the influence function of T is:

$$\varphi(x) = (x^2 - m_2) - 2\mu_1(x - \mu_1).$$



Example 3.4 (Third Central Moment). The 3rd central moment functional is:

$$T(F) = \int \{x - T_1(F)\}^3 dF(x).$$

Upon expansion:

$$T(F) = T_3(F) - 3T_2(F)T_1(F) + 2T_1^3(F)$$

By linearity, the product and chain rules, the influence function is:

$$\varphi(x) = (x^3 - \mu_3) - 3\{(x^2 - m_2)\mu_1 + m_2(x - \mu_1)\} + 6\mu_1^2(x - \mu_1).$$



3.3 von Mises Expansion

Definition 3.2 (Serfling 1980, section 6.2). Suppose T is a statistical functional. The **von Mises expansion** of order m is:

$$T(\mathbb{F}_n) - T(F) = V_{n,m} + R_{n,m}, \quad (3.2)$$

where $V_{n,m}$ is given by:

$$V_{n,m} = \sum_{k=1}^m \frac{1}{k!} \partial^k T(F; \mathbb{F}_n - F)$$

The *remainder* is expressible either as $R_{n,m} = \{T(\mathbb{F}_n) - T(F)\} - V_{n,m}$, or as:

$$R_{n,m} = \frac{1}{(m+1)!} \left[\frac{\partial^{m+1}}{\partial \epsilon^{m+1}} T\{F + \epsilon(\mathbb{F}_n - F)\} \right]_{\epsilon=\epsilon^*}$$

for $\epsilon^* \in [0, 1]$. To validate the von Mises expansion (3.2), it is necessary to show that the scaled remainder is asymptotically negligible:

$$n^{m/2} R_{n,m} \xrightarrow{p} 0.$$

■

3.4 Asymptotic Normality

Proposition 3.2. Suppose $(X_i) \stackrel{\text{iid}}{\sim} F$ and T is a statistical functional that admits an *influence function expansion* of the form:

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_F(X_i) + \sqrt{n}R_n,$$

where $\mathbb{E}\{\varphi_F(X)\} = 0$ and $\sqrt{n}R_n \xrightarrow{p} 0$. Then:

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} \xrightarrow{\mathcal{L}} N(0, \gamma_F^2), \quad (3.3)$$

where $\gamma_F^2 = \mathbb{E}\{\varphi_F^2(X)\}$.

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Proof. Since T admits an influence function:

$$\begin{aligned} T(\mathbb{F}_n) - T(F) &= \int \varphi_F(x) d\mathbb{F}_n(x) + R_n \\ &= \int \varphi_F(x) d\left\{ \frac{1}{n} \sum_{i=1}^n H_{X_i}(x) \right\} + R_n = \frac{1}{n} \sum_{i=1}^n \varphi_F(X_i) + R_n. \end{aligned}$$

Rescaling by $\sqrt{n}$ gives:

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_F(X_i) + \sqrt{n}R_n.$$

Since $\mathbb{E}\{\varphi_F(X_i)\} = 0$, by the ordinary CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_F(X_i) \xrightarrow{\mathcal{L}} N(0, \gamma_F^2).$$

By hypothesis, the scaled remainder is asymptotically negligible $\sqrt{n}R_n \xrightarrow{p} 0$. The conclusion follows from Slutsky's theorem. ■

Remark 3.1 (Lehmann 1999, section 6.3). Consistency and asymptotic normality of $T(\mathbb{F})$ as an estimator of $T(F)$ may be established via the following steps:

- i. Derive the influence function for T :

$$\varphi_F(x) = \lim_{\epsilon \downarrow 0} \left[\frac{\partial}{\partial \epsilon} T\{F + \epsilon(H_x - F)\} \right].$$

- ii. Verify that the influence function is unbiased:

$$\int \varphi_F(x) dF(x) = 0.$$

- iii. Verify that the remainder is asymptotically negligible:

$$\sqrt{n}R_n = \sqrt{n} \left\{ T(\mathbb{F}_n) - T(F) - \frac{1}{n} \sum_{i=1}^n \varphi_F(X_i) \right\} \xrightarrow{p} 0.$$

- iv. Determine the asymptotic variance:

$$\gamma_F^2 = \mathbb{E}\{\varphi_F^2(X)\} = \int \varphi_F^2(x) dF(x).$$

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3.5 Examples

Example 3.5 (Cramer von Mises Distance). Recall from a previous example that the Gateaux derivative of the Cramer von Mises functional is:

$$\partial T(F; G - F) = 2 \int (F - F_0)(G - F) dF_0.$$

Therefore, the candidate influence function is:

$$\varphi_F(x) = 2 \int (F - F_0)(H_x - F) dF_0.$$

The influence function is unbiased since:

$$\mathbb{E}\{\varphi_F(X)\} = 2 \int (F - F_0)\{E(H_X) - F\} dF_0 = 2 \int (F - F_0)(F - F) dF_0 = 0.$$

The remainder takes the form:

$$\begin{aligned} R_n &= T(\mathbb{F}_n) - T(F) - \frac{2}{n} \sum_{i=1}^n \int (F - F_0)(H_{X_i} - F) dF_0 \\ &= \int (\mathbb{F}_n - F_0)^2 dF_0 - \int (F - F_0)^2 dF_0 - 2 \int (F - F_0)(\mathbb{F}_n - F) dF_0 \\ &= \int \{(\mathbb{F}_n - F_0)^2 - (F - F_0)^2 - 2(F - F_0)(\mathbb{F}_n - F)\} dF_0 \\ &= \int \{\mathbb{F}_n - 2\mathbb{F}_n F + F^2\} dF_0 = \int (\mathbb{F}_n - F)^2 dF_0. \end{aligned}$$

The remainder is bounded above by:

$$|R_n| \leq \sup_{x \in \mathbb{R}} \{\mathbb{F}_n(x) - F(x)\}^2.$$

By the Dvoretzky, Kiefer, Wolfowitz inequality:

$$\sqrt{n} \left\{ \sup_{x \in \mathbb{R}} |\mathbb{F}_n(x) - F(x)| \right\} = \mathcal{O}_p(1),$$

therefore:

$$\sqrt{n} |R_n| \leq \frac{1}{\sqrt{n}} \cdot n \cdot \sup_{x \in \mathbb{R}} \{\mathbb{F}_n(x) - F(x)\}^2 = \mathcal{O}(n^{-1/2}) \xrightarrow{p} 0.$$

This establishes the influence function expansion for the Cramer von Mises functional:

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} = \frac{2}{\sqrt{n}} \sum_{i=1}^n \int (F - F_0)(H_{X_i} - F) dF_0 + \mathcal{O}_p(n^{-1/2}).$$

Now,

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} \xrightarrow{\mathcal{L}} N(0, \gamma_F^2),$$

with asymptotic variance:

$$\gamma_F^2 = \mathbb{E}\{\varphi_F^2(X)\} = 4E_F \left\{ \int (F - F_0)(H_X - F) dF_0 \right\}^2.$$



Example 3.6 (Sample Quantile). Consider the p th sample quantile:

$$t_p = T(F) = F^{-1}(p).$$

Towards obtaining the influence function, let $t_{p,\epsilon} = T(F_{x,\epsilon}) = F_{x,\epsilon}^{-1}(p)$, then:

$$p = F_{x,\epsilon}(t_{p,\epsilon}) = (1 - \epsilon)F(t_{p,\epsilon}) + \epsilon H_x(t_{p,\epsilon}).$$

Taking the implicit derivative with respect to ϵ :

$$0 = -F(t_{p,\epsilon}) + (1 - \epsilon)f(t_{p,\epsilon})\frac{\partial t_{p,\epsilon}}{\partial \epsilon} + H_x(t_{p,\epsilon}) + \epsilon \delta_x(t_{p,\epsilon})\frac{\partial t_{p,\epsilon}}{\partial \epsilon}$$

In the limit as $\epsilon \downarrow 0$:

$$0 = -F(t_p) + f(t_p) \left\{ \frac{\partial t_{p,\epsilon}}{\partial \epsilon} \right\}_{\epsilon=0^+} + H_x(t_p).$$

Identify $\{\partial_\epsilon t_{p,\epsilon}\}_{\epsilon=0^+}$ as the influence function, and rearrange to obtain:

$$\varphi_F(x) = \frac{F(t_p) - H_x(t_p)}{f(t_p)} = \frac{p - \mathbb{I}(x \leq t_p)}{f(t_p)}.$$

The influence function is unbiased:

$$\mathbb{E}\{\varphi_F(X)\} = \frac{p - F(t_p)}{f(t_p)} = \frac{p - p}{f(t_p)} = 0.$$

From Serfling (1980), theorem 2.5.1:

$$\sqrt{n} \left(T(\mathbb{F}_n) - T(F) - \frac{1}{n} \sum_{i=1}^n \frac{p - \mathbb{I}(X_i \leq t_p)}{f(t_p)} \right) \xrightarrow{p} 0.$$

Therefore:

$$\sqrt{n}\{T(\mathbb{F}_n) - T(F)\} \xrightarrow{\mathcal{L}} N(0, \gamma_F^2),$$

where the asymptotic variance is:

$$\gamma_F^2 = \mathbb{V}\{\varphi_F(X)\} = \frac{F(t_p)\{1 - F(t_p)\}}{f^2(t_p)} = \frac{p(1 - p)}{f^2(t_p)}.$$



Example 3.7 (M-Estimation). Suppose $(X_i) \stackrel{\text{iid}}{\sim} F$, and consider an M -estimator $\hat{\theta}_n = T(\mathbb{F}_n)$, defined implicitly by the relation:

$$0 = \frac{1}{n} \sum_{i=1}^n \psi(X_i; \hat{\theta}_n) = \int \psi\{u; T(\mathbb{F}_n)\} d\mathbb{F}_n(u),$$

where ψ is the *estimating equation*. Towards finding the influence function, consider evaluation of the estimating relation at $F_{x,\epsilon}$:

$$\begin{aligned} 0 &= \int \psi\{u; T(F_{x,\epsilon})\} dF_{x,\epsilon}(u) = \int \psi\{u; T(F_{x,\epsilon})\} d\{(1-\epsilon)F(u) + \epsilon H_x(u)\} \\ &= (1-\epsilon) \int \psi\{u; T(F_{x,\epsilon})\} dF(u) + \epsilon \psi\{x; T(F_{x,\epsilon})\} \end{aligned}$$

Taking the implicit derivative with respect to ϵ :

$$\begin{aligned} 0 &= - \int \psi\{u; T(F_{x,\epsilon})\} dF(u) + (1-\epsilon) \int \dot{\psi}\{u; T(F_{x,\epsilon})\} \frac{\partial T(F_{x,\epsilon})}{\partial \epsilon} dF(u) \\ &\quad + \psi\{x; T(F_{x,\epsilon})\} + \epsilon \dot{\psi}\{x; T(F_{x,\epsilon})\} \frac{\partial T(F_{x,\epsilon})}{\partial \epsilon}. \end{aligned}$$

In the limit as $\epsilon \downarrow 0$:

$$\begin{aligned} 0 &= - \int \psi\{u; T(F)\} dF(u) + \int \dot{\psi}\{u; T(F_{x,\epsilon})\} dF(u) \cdot \left\{ \frac{\partial T(F_{x,\epsilon})}{\partial \epsilon} \right\}_{\epsilon=0^+} + \psi\{x; T(F)\} \\ &= -E_F\{\psi(X; \theta)\} + E_F\{\dot{\psi}(X; \theta)\} \cdot \varphi_F(x) + \psi(x; \theta) \end{aligned}$$

Upon rearranging:

$$\varphi_F(x) = - \frac{\psi(x; \theta) - E_F\{\psi(X; \theta)\}}{E_F\{\dot{\psi}(X; \theta)\}}.$$

Under the regularity conditions for M -estimation the remainder is asymptotically negligible, and:

$$\sqrt{n}\{\hat{\theta}_n - \theta\} \xrightarrow{\mathcal{L}} N(0, \gamma_F^2),$$

with asymptotic variance:

$$\mathbb{V}\{\varphi_F(X)\} = \frac{\mathbb{V}\{\psi(X; \theta)\}}{E_F^2\{\dot{\psi}(X; \theta)\}}.$$

In the case of maximum likelihood estimation, $\psi(x; \theta) = \dot{\ell}(x; \theta)$, $\ell(x; \theta) = \ln f(x; \theta)$, is the *score equation*, which is unbiased $\mathbb{E}\{\psi(X; \theta)\} = \mathbb{E}\{\dot{\ell}(X; \theta)\} = 0$.

The negative expectation of $\dot{\psi}(X; \theta)$ is the Fisher information:

$$-E_F\{\dot{\psi}(X; \theta)\} = -E_F\{\ddot{\ell}(X; \theta)\} = \mathcal{I}(\theta).$$

The influence function takes the form:

$$\varphi_F(x) = -\mathcal{I}^{-1}(\theta)\dot{\ell}(x; \theta),$$

and the asymptotic variance is:

$$\gamma_F^2 = \frac{\mathbb{V}\{\dot{\ell}(X; \theta)\}}{\mathcal{I}^2(\theta)} = \frac{\mathcal{I}(\theta)}{\mathcal{I}^2(\theta)} = \mathcal{I}^{-1}(\theta).$$



Example 3.8 (V-Estimation). Consider a V -estimator of order $m = 2$ with symmetric kernel function $h(X_1, X_2)$:

$$T(F) = \int h(x_1, x_2) dF(x_1) dF(x_2).$$

Towards finding the influence function, consider the evaluation at $F_{x,\epsilon}$:

$$\begin{aligned} T(F_{x,\epsilon}) &= \int h(x_1, x_2) d\{(1-\epsilon)F(x_1) + \epsilon H_x(x_1)\} d\{(1-\epsilon)F(x_2) + \epsilon H_x(x_2)\} \\ &= (1-\epsilon)^2 \int h(x_1, x_2) dF(x_1) dF(x_2) + (1-\epsilon)\epsilon \int h(x_1, x) dF(x_1) \\ &\quad + \epsilon(1-\epsilon) \int h(x, x_2) dF(x_2) + \epsilon^2 h(x, x). \end{aligned}$$

Recall that $\theta = E_F\{h(X_1, X_2)\}$, and that $h_1(x) = \mathbb{E}\{h(X_1, X_2)|X_1 = x\}$ denotes the first projection. The functional at $F_{x,\epsilon}$ is expressible as:

$$T(F_{x,\epsilon}) = (1-\epsilon)^2\theta + 2\epsilon(1-\epsilon)h_1(x) + \epsilon^2 h(x, x).$$

Taking the derivative with respect to ϵ :

$$\frac{\partial T(F_{x,\epsilon})}{\partial \epsilon} = -2(1-\epsilon)\theta + 2(1-\epsilon)h_1(x) - 2\epsilon h_1(x) + 2\epsilon h(x, x).$$

In the limit as $\epsilon \downarrow 0$:

$$\varphi_F(x) = 2\{h_1(x) - \theta\}$$

The remainder takes the form:

$$R_n = \frac{1}{n^2} \sum_{j=1}^n \sum_{i=1}^n \{h(X_i, X_j) - \theta\} - \frac{2}{n} \sum_{i=1}^n \{h_1(X_i) - \theta\}$$

Supposing $\mathbb{V}\{h(X, X)\} < \infty$, it can be shown that $n\text{Var}(R_n) \rightarrow 0$ (see Serfling 1980, 6.3.2), which implies $\sqrt{n}R_n = o_p(1)$. Therefore:

$$\sqrt{n}\{\hat{\theta}_n - \theta\} \xrightarrow{\mathcal{L}} N(0, \gamma_F^2),$$

where the asymptotic variance is:

$$\mathbb{V}\{\varphi_F(X)\} = 4\mathbb{V}\{h_1(X)\} = 4\zeta_1^2.$$



Jackknife

4.1 Definition

Definition 4.1. Suppose $(X_i) \stackrel{\text{iid}}{\sim} F$, and let $\hat{\theta}_n = t(x_1, \dots, x_n)$ denote an estimator. Denote by $\hat{\theta}_{(-i)} = t(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ the estimate obtained by omitting the i th observation. Define the i th **pseudo-value** as:

$$\tilde{\theta}_i = n\hat{\theta}_n - (n-1)\hat{\theta}_{(-i)}.$$

The jackknife estimator of θ is:

$$\hat{\theta}_J = \frac{1}{n} \sum_{i=1}^n \tilde{\theta}_i.$$

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4.2 Empirical Influence Function

Proposition 4.1. Define $\mathbb{F}_{(-i)}$ as the empirical distribution function obtained by omitting the i th observation. $\mathbb{F}_{(-i)}$ is related to the overall distribution function $\mathbb{F}_n$ via:

$$\mathbb{F}_{(-i)} = \mathbb{F}_n + \epsilon_n(H_{X_i} - \mathbb{F}_n).$$

Thus, $\mathbb{F}_{(-i)}$ may be regarded as an ϵ -contaminated distribution (2.3). ◆

Proof. Expressing $\mathbb{F}_n$ in terms of $\mathbb{F}_{(-i)}$:

$$\mathbb{F}_n = \frac{1}{n}H_{X_i} + \frac{1}{n} \sum_{j \neq i}^n H_{X_j} = \frac{1}{n}H_{X_i} + \frac{n-1}{n} \frac{1}{n-1} \sum_{j \neq i} \delta_{X_j} = \frac{1}{n}H_{X_i} + \frac{n-1}{n} \mathbb{F}_{(-i)}.$$

Solving for $\mathbb{F}_{(-i)}$:

$$\mathbb{F}_{(-i)} = \frac{n}{n-1} \mathbb{F}_n - \frac{1}{n-1} H_{X_i} = \left(1 + \frac{1}{n-1}\right) \mathbb{F}_n - \frac{1}{n-1} H_{X_i}.$$

Let $\epsilon_n = -(n-1)^{-1}$, then:

$$\mathbb{F}_{(-i)} = \mathbb{F}_n - \epsilon_n \mathbb{F}_n + \epsilon_n H_{X_i} = \mathbb{F}_n + \epsilon_n (H_{X_i} - \mathbb{F}_n).$$

■

Proposition 4.2. The difference between the i th *pseudo-value* $\tilde{\theta}_i$ and the overall sample estimator $\hat{\theta}_n$ is an empirical approximation to the influence function at X_i :

$$\tilde{\theta}_i - \hat{\theta}_n = \hat{\varphi}_F(X_i).$$

◆

Proof. Consider the i th pseudo-value:

$$\begin{aligned}\tilde{\theta}_i &= n\hat{\theta}_n - (n-1)\hat{\theta}_{(-i)} = \hat{\theta}_n - (n-1)(\hat{\theta}_{(-i)} - \hat{\theta}_n), \\ \tilde{\theta}_i - \hat{\theta}_n &= \frac{\hat{\theta}_{(-i)} - \hat{\theta}_n}{-(n-1)^{-1}} = \frac{\hat{\theta}_{(-i)} - \hat{\theta}_n}{\epsilon_n}.\end{aligned}$$

Writing $\theta = T(F)$ as a statistical functional:

$$\tilde{\theta}_i - \hat{\theta}_n = \epsilon_n^{-1} \{T(\mathbb{F}_{(-i)}) - T(\mathbb{F}_n)\} = \epsilon_n^{-1} [T\{\mathbb{F}_n + \epsilon_n(H_{X_i} - \mathbb{F}_n)\} - T(\mathbb{F}_n)].$$

Comparison to the Gateaux derivative (2.4) suggests that for $n \rightarrow \infty$:

$$\tilde{\theta}_i - \hat{\theta}_n \approx \varphi_F(X_i).$$

■

Example 4.1 (Jackknife Variance Estimator). Suppose $\hat{\theta}_n = T(\mathbb{F}_n)$ admits an influence function expansion:

$$\sqrt{n}(\hat{\theta}_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_F(X_i) + o_p(1).$$

The asymptotic variance of $\sqrt{n}(\hat{\theta}_n - \theta)$ is $\mathbb{V}\{\varphi_F(X)\}$, which is estimated by:

$$\hat{V}\{\varphi_F(X)\} \doteq \frac{1}{n} \sum_{i=1}^n \varphi_F^2(X_i).$$

Approximating the true influence function φ_F by the empirical influence of X_i :

$$\frac{1}{n} \sum_{i=1}^n \varphi_F^2(X_i) \approx \frac{1}{n} \sum_{i=1}^n \hat{\varphi}_F^2(X_i) = \frac{1}{n} \sum_{i=1}^n (\tilde{\theta}_i - \hat{\theta}_n)^2.$$

This approximation motivates the jackknife variance estimator:

$$\hat{V}_J\{\sqrt{n}(\hat{\theta}_n - \theta)\} = \frac{1}{n-1} \sum_{i=1}^n (\tilde{\theta}_i - \hat{\theta}_n)^2.$$

Note that this is an estimate of $\mathbb{V}\{\sqrt{n}(\hat{\theta}_n - \theta)\}$. The jackknife estimate of $\text{Var}(\hat{\theta}_n)$ is:

$$\hat{V}_J(\hat{\theta}_n) = \frac{1}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (\tilde{\theta}_i - \hat{\theta}_n)^2.$$

Expressed in terms of the leave-1-out estimates:

$$\hat{V}_J(\hat{\theta}_n) = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \hat{\theta}_n)^2.$$

♠

References

References for influence functions:

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